

**4.10 Solution:** We can solve this problem by expanding the scalar potential in Legendre polynomials, but need to be careful that the Legendre polynomials involved are restricted to the half domain. Therefore, to find the expansion coefficients, we need to use the fact that Legendre polynomials of even or odd orders form orthonormal complete base on the half domain. We will not pursue this approach, but rather will rely on symmetry to simplify the problem.

(a) From symmetry consideration, the electric field must be radial. Let the electric fields in the left and the right hemisphere are  $E_L$  and  $E_R$ , respectively. At the interface between air and the dielectric, the electric fields are parallel to the interface, and by the requirement of continuity of tangential components of the electric field, we must have  $E_L = E_R = E$ . Using Gauss law, we have

$$\oint \mathbf{D} \cdot \mathbf{n} da = Q,$$

where the integral can be easily evaluated as  $2\pi r^2(\epsilon + \epsilon_0)E$ . Therefore, the electric field between the spheres is

$$E = \frac{Q}{2\pi(\epsilon + \epsilon_0)r^2}.$$

(b) On the inner sphere, the charge density is given by  $\sigma = \mathbf{D} \cdot \mathbf{n}$ , where  $\mathbf{n}$  is a unit vector pointing outward. Since  $\mathbf{D}$  is also radial, we then have  $\sigma = D$ . Therefore,

$$\sigma_L = \epsilon_0 E(r = a) = \frac{\epsilon_0}{\epsilon + \epsilon_0} \frac{Q}{2\pi a^2}, \quad \sigma_R = \epsilon E(r = a) = \frac{\epsilon}{\epsilon + \epsilon_0} \frac{Q}{2\pi a^2}.$$

(c) Clearly,

$$\sigma_P = \frac{\epsilon - \epsilon_0}{\epsilon + \epsilon_0} \frac{Q}{2\pi a^2}$$